

CONVERGENCE FELLOWSHIP PROGRAM

Lead, Own, Share

Sovereign Wealth Funds for Transformative AI

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Abstract

This paper proposes sovereign wealth funds (SWFs) as a foundational policy instrument for states to navigate the economic and strategic challenges posed by transformative artificial intelligence (TAI). I explore the rationale for public investment in AI-related assets, drawing on analogies from resource governance and historical state-led development. I evaluate how different types of SWFs—strategic, stabilization, and savings funds—could be adapted to invest in compute infrastructure, upstream supply chains, and equity in frontier AI labs. My analysis identifies key pathways through which public ownership can shape TAI development, mitigate economic volatility, and distribute AI-derived wealth more equitably. I assess the design tradeoffs of ownership models, from minority stakes to majority control, and highlight political and institutional constraints on implementation. By comparing SWFs’ potential to that of laissez-faire or purely regulatory approaches, I argue that SWFs offer a flexible, scalable tool for democratic states to retain strategic influence over TAI while preserving fiscal legitimacy. I conclude by outlining scenarios under which SWFs may succeed or fail, providing a structured framework for policymakers to evaluate this institutional lever in managing the AI transition.

Executive Summary

In this paper, I propose that **sovereign wealth funds (SWFs)** should be considered a foundational governance tool for managing strategic and economic challenges posed by **transformative artificial intelligence (TAI)**. SWFs—state-owned investment vehicles—can be adapted to serve three critical roles in a TAI context:

1. **Lead:** Supporting national leadership in frontier AI development through strategic investment in upstream infrastructure and key AI firms.
2. **Own:** Acquiring equity in frontier AI companies to secure public influence over value capture.
3. **Share:** Distributing the economic returns from TAI in ways that preserve public legitimacy, stabilize the economy, and prevent elite entrenchment.

I evaluate SWFs as a flexible institutional lever, especially for democratic states seeking durable and non-coercive ways to shape AI trajectories.

This summary outlines the key conclusions of my paper. For the reasoning and analysis behind each, see the corresponding sections.

Why SWFs for AI Governance?

AI is increasingly viewed as a likely driver of future economic and geopolitical power. To ensure its development aligns with democratic interests, I propose that:

- **SWFs can secure long-horizon public influence** over upstream AI infrastructure (e.g. chips, energy) and downstream AI companies.
- **Without public ownership**, benefits from TAI are likely to concentrate among a small set of unaccountable actors.
- **Public ownership via SWFs is more resilient than regulation alone**, which may face delay, rollback, or jurisdictional limits.

Proposed Uses

Enabling U.S. AI Leadership

I propose that a U.S. strategic fund could:

- Invest in **domestic semiconductor manufacturing**, especially where the U.S. currently lacks cutting-edge capacity.
- Co-invest in **allied semiconductor ecosystems** to increase redundancy and resilience in upstream chip inputs.
- Support **energy infrastructure buildout** to meet the immense power

demands of AI training.

- Fund **security-critical infrastructure** (e.g., secure-by-design hardware) before commercial incentives emerge.
- Acquire **early minority stakes in frontier AI companies** to gain access, governance rights, and strategic veto power without full control.

Ownership Pathways

I evaluate a spectrum of state ownership models:

1. **No stake:** maximizes private control; risks misalignment, elite capture, or regulatory failure.
2. **Minority stake:** allows targeted influence, often via side letters, board rights, or golden shares.
3. **Majority stake/nationalization:** allows full control but poses high political and financial barriers.

I conclude that minority equity positions are the most feasible and strategically attractive lever for influence.

Public Management of AI-Derived Wealth

- I propose using **savings funds** to distribute universal basic dividends, providing citizens a share of AI-derived wealth.
- **Stabilization funds** can smooth tax revenues and labor shocks caused by rapid automation by supporting displaced workers with income supplements and retraining, deploying automatic stabilizers, and maintaining core public services amid economic volatility.

Design Principles

- SWFs could be **explicitly mandated** to serve public benefit, with transparent objectives.
- **Governance risks** (e.g., elite capture, opacity, political misuse) could be addressed via legal structures, reporting requirements, and independent oversight.
- The erosion of public trust can be mitigated by:
 - Avoiding full nationalization unless necessary.
 - Embedding social objectives contractually or in the fund's mandate.

Limitations and Risks

I identify six key limitations and risks that could undermine the effectiveness of SWFs for TAI:

- **Institutional fragility:** Rapid AI progress may overwhelm or erode state capacity for funds to act as intended.

- **Speculative uncertainty:** If TAI fails to materialize, fund performance may underdeliver, though strategic investments (e.g. energy) may still yield benefits.
- **Arms race escalation:** State investment in AI could accelerate geopolitical rivalry and reduce cooperation incentives.
- **Governance risks:** SWFs may be captured by elite interests or used for political patronage, undermining public benefit.
- **Investment constraints:** States may face fiscal, legal, or political barriers to investing in frontier AI companies, limiting their ability to secure public ownership.
- **Global exclusion:** States unable to invest in TAI may be excluded from its benefits, reinforcing global inequalities.

Conclusion

If TAI becomes a dominant source of economic value in the near future, then ownership and governance structures will shape not only who captures that value, but also how it impacts economic stability, inequality, and political authority. SWFs offer a uniquely adaptive tool for helping states lead AI development, exert meaningful public ownership, and share the benefits of TAI more equitably.

Introduction

Transformative artificial intelligence (TAI)¹ may redefine global power structures, national economies, and the foundations of labor and wealth. States that achieve TAI first can shape global norms and wield disproportionate power through its application. This dynamic raises significant concerns about who governs the trajectory of advanced AI and under what values. Ensuring development aligns with public accountability and democratic principles may require sustained influence by states committed to transparency and broad-based benefits. While the United States holds a relative advantage in AI talent and compute infrastructure, its continued leadership is not guaranteed. Without strategic public investment, states with more centralized coordination or greater capacity for long-term resource mobilization could gain a competitive edge, potentially reducing the scope for democratic influence over advanced AI's direction.

Even if democratic states maintain leadership in TAI, that does not guarantee equitable or stable outcomes, as most economic gains may concentrate in the hands of a narrow set of actors who control TAI. AI systems are already automating high-skill cognitive tasks [2]. If this trend continues, labor income could decline, tax bases tied to human work could erode, and political legitimacy could fray. A public stake in AI assets may help mitigate these risks, but ownership alone is insufficient. State control over TAI-related capital could be paired with institutions capable of stabilizing the economy and distributing gains in ways that preserve democratic legitimacy and ensure broad-based economic participation.

Despite growing concern over the societal impacts of TAI [1], there remains little concrete policy research on how states might shape long-term economic outcomes in a post-TAI world. In particular, there is a lack of proposals that address how governments can proactively capture and distribute value in an economy increasingly driven by automated intelligence. This paper examines the potential of sovereign wealth funds (SWFs), state-owned investment vehicles, as a promising institutional tool for shaping the governance of TAI. It argues that TAI may become the future's defining strategic and economic resource and that SWFs offer a flexible mechanism for states to influence its development, ownership, and distribution.

This paper adopts a conceptual and policy-analytic methodology, drawing on political economy theory, historical analogies from state-led development, and empirical data on frontier AI capabilities, investment patterns, and infrastructure requirements. It does not employ formal modeling or econometric analysis but develops a structured argument through scenario analysis and case studies.

¹ TAI is defined as “any AI technology or application with potential to lead to practically irreversible change that is broad enough to impact most important aspects of life and society” [1].

Recent political developments heighten the relevance of this inquiry. On February 3, 2025, US President Donald Trump signed Executive Order 14196, initiating a plan to establish a national SWF to promote fiscal sustainability, national economic security, and US strategic leadership [2]. In parallel, the concept of “sovereign AI” has gained traction, referring to national efforts to secure control over AI infrastructure, talent, and data to maintain technological autonomy and align development with public values [3]. These developments underscore the timeliness of evaluating how SWFs could help states navigate the strategic and distributive challenges TAI poses. While this paper gives particular attention to the US context, its broader ambition is to evaluate how SWFs might serve any state seeking to steer the societal impacts of TAI.

To structure the analysis, this paper proceeds in seven sections.

Chapter 2 provides a brief primer on SWFs, focusing on strategic, stabilization, and savings funds.

Chapter 3 explores the geopolitical stakes of leading in frontier AI development. It proposes a US strategic fund model to invest in critical inputs, domestic and allied chip production, energy production, security infrastructure, and equity stakes in leading AI companies.

Chapter 4 addresses the public ownership of TAI. If AI increasingly automates large segments of the economy, who captures the resulting value becomes a central concern. This chapter evaluates various state ownership models, from no stake to majority control, and analyzes their implications for public influence over TAI.

Chapter 5 addresses the public management of AI-derived wealth. Even if the state acquires AI-related assets, societal benefit is not guaranteed. This chapter contrasts a laissez-faire approach with models for distributing public returns. It explores how savings and stabilization funds could provide economic security in a post-labor economy.

Chapter 6 provides a broader discussion of institutional and policy challenges, including the persistence of state institutions amid rapid change, the uncertainty regarding the emergence of TAI, the risk of escalating arms race dynamics, the governance risks of SWFs, practical barriers to investing in frontier AI companies, and the issue of global benefit-sharing.

Chapter 7 concludes by reflecting on SWFs’ promise and limitations as tools for shaping TAI’s development, ownership, and distribution.

Primer on Sovereign Wealth Funds

SWFs are state-owned investment vehicles established to manage national assets in support of long-term financial and macroeconomic objectives [4]. As of 2025, these funds collectively manage over \$9 trillion in assets [5].

According to the Santiago Principles—internationally recognized guidelines for the governance of SWFs—such funds must be wholly owned by a central or subnational government, invest at least part of their portfolio in foreign financial assets, and pursue primarily financial objectives [6].

The concept of SWFs dates back to the 1950s, with the Kuwait Investment Authority being the first established fund [7]. Historically, SWFs were created by commodity-rich countries to manage budget surpluses and buffer their economies from the volatility of resource revenues. Over time, however, the role of SWFs has evolved. Today, they are used to manage excess revenues and achieve a range of goals.

SWFs can be categorized into three primary types: strategic (or development) funds, stabilization funds, and savings funds, though many funds operate with hybrid objectives [8].

Strategic funds, also known as development funds, focus on national economic development [4]. They invest in domestic infrastructure, strategic industries, startups, and long-term ventures, often operating similarly to venture capital investors [9].

Stabilization funds aim to cushion government budgets against revenue volatility, particularly in countries dependent on commodity exports, and invest primarily in highly liquid and low-risk assets, allowing governments to save during periods of high commodity prices and draw upon reserves during downturns [4][10].

Savings funds, designed with intergenerational equity in mind, seek to transform finite or volatile revenues, such as those from oil or gas, into diversified, long-term financial assets, and accept higher levels of investment risk to achieve greater returns for future generations [4][11].

Leadership in Frontier AI Development

Geopolitics of AI Leadership

Leadership in frontier AI development may shape global economic and military dynamics and the trajectory of societal values embedded in future technologies. As we approach TAI, the political context in which these technologies are developed becomes increasingly consequential. There is a compelling case that frontier AI development should occur within democratic rather than authoritarian states. Democracies may be more likely to align technological progress with broadly beneficial, accountable, and transparent outcomes [12], while authoritarian governments have already leveraged AI to expand surveillance, suppress dissent, and entrench power [13][14], and may continue to do so.

Authoritarianism, characterized by concentrated authority and limited freedoms [15], could be augmented by AI. For example, China is integrating AI not merely for modernization but to reinforce the legitimacy and control of the Chinese Communist Party through economic performance and ideological stability [14]. AI can be seen as an “enabling technology” that can shift the global balance of military and economic power, especially depending on how its capabilities are diffused [16]. This context heightens the importance of who leads in frontier AI development. Beyond economic competitiveness, the entity that pulls ahead in AI capabilities may gain a “decisive strategic advantage”: a scenario in which a single actor, possibly an AI itself, consolidates unassailable global power [17]. This could lock in the values and priorities of whoever controls the system, making democratic leadership preferable and potentially existentially important.

As the United States and China lead frontier AI development [18], the geopolitical implications of AI leadership are most directly shaped by their respective trajectories. While the degree of democratic governance varies, recent assessments classify the United States as a liberal democracy and China as a closed autocracy [19][20]. Given the preference for democratic stewardship of AI, it is prudent to concentrate efforts on strengthening frontier AI development within the US context. This section will focus on how the United States can maintain leadership in frontier AI development.

While the United States still leads in total compute capacity, China is rapidly closing the gap in AI model performance [21]. Frontier AI development hinges on scaling effective compute, the key driver of AI capabilities, but this requires massive investment in hardware, software, and R&D, and is constrained by

supply chain bottlenecks, capital-intensive infrastructure, and adjustment frictions that make rapid expansion disproportionately costly [22].

Technological leadership depends not just on developing new technologies but on deploying them at scale, an area where China is often seen as strong [23], though recent analyses offer more mixed assessments [24]. Finally, security is important: the theft of model weights, architectures, or training data by hostile states or actors could result in the uncontrolled spread of powerful AI systems. Model weight theft grants unrestricted access and circumvents safety controls, with current US defenses inadequate against state-level threats [25].

China and the United States have explicitly recognized AI leadership as a strategic national priority. China announced a strategy to achieve AI dominance by 2030 [26], and the United States has framed AI leadership as essential to economic competitiveness and national security, mandating a push to strengthen domestic leadership [27]. As both nations mobilize to secure long-term advantages in AI, new forms of state-backed investment are becoming increasingly relevant. While existing entities like the Defense Advanced Research Projects Agency, In-Q-Tel, and the Office of Strategic Capital support targeted innovation in the United States, a SWF has been proposed to complement their narrower scope and address gaps left by private markets through broader strategic and economic investment [28]. Unlike traditional SWFs, which focus on financial returns from diversified global assets, strategic funds pursue strategic national objectives through targeted domestic investments [29]. To keep frontier AI leadership anchored in a democratic country, the United States could establish a strategic fund that provides long-term capital to advance frontier AI development and secure its development pipeline.

The following section outlines how a US strategic fund could advance AI leadership by securing domestic semiconductor production, co-investing with allies, expanding energy production, supporting security infrastructure, and taking strategic equity in frontier AI companies.

Securing US AI Leadership

Domestic Semiconductor Production

A strategic fund could directly finance frontier AI development and secure upstream components of the AI ecosystem, particularly the semiconductor supply chain. Chips are foundational to AI model training and deployment, but the United States remains strategically exposed in several critical segments of chip production.

While the United States leads in research and development, chip design, and electronic design automation, contributing roughly 39% of global semiconductor value added, it lacks domestic capacity in key production stages [30]. Most concerning, the United States does not operate cutting-edge pure-play foundries, which manufacture chips for other companies, leaving it reliant

on firms based in Taiwan and South Korea [31]. Further downstream, over 95% of assembly, testing, and packaging (ATP) occurs in the Indo-Pacific, a region vulnerable to conflict or coercion, especially amid rising US-China tensions [32]. These dependencies are logistical and geopolitical chokepoints. A strategic fund could help close these gaps through domestic investment and selective foreign acquisition. Domestically, it could fund the expansion of critical manufacturing and ATP capacity, helping insulate the AI pipeline from foreign shocks. Internationally, it could acquire minority or strategic equity in foreign firms controlling key supply nodes to diversify risk and gradually build industrial capability through what scholars call “long-term learning” [33]. There is precedent for this model. Mubadala, the UAE's SWF, used a series of strategic semiconductor investments, including stakes in AMD and the acquisition of GlobalFoundries, to build national competitiveness in chips [34]. China's SWFs, especially the China Investment Corporation (CIC), have acquired stakes in foreign and domestic semiconductor firms to expand capacity, accelerate technology transfer, and ensure resilience in times of crisis [35][36]. During the early COVID-19 pandemic, these stakes enabled China to prioritize domestic needs through influence over global distribution networks. Such strategies also offer leverage. Equity positions, particularly those conferring board rights or majority ownership, could allow states to influence corporate governance, redirect capital, and incentivize foreign firms to locate R&D or fabrication capacity domestically. Although politically sensitive, this strategy has been selectively deployed by US allies such as Taiwan, South Korea, and especially Singapore, which has directly acquired foreign corporations to promote technological innovation in domestic industries [37].

Allied Semiconductor Production

While political instability often dominates discussion of supply chain risk, sourcing critical technologies from abroad exposes domestic industries to broader vulnerabilities, including natural disasters, trade restrictions, and transportation disruptions [38]. These risks are particularly acute in the semiconductor industry, where the production of key components remains highly geographically concentrated. For example, the most advanced photolithography tools, such as extreme ultraviolet scanners and argon fluoride immersion scanners, are only manufactured in the Netherlands and Japan, respectively [38]. A disruption in either country could halt the global production of cutting-edge chips. Instead of relocating production entirely, the United States might benefit from helping these allies scale up and diversify their capabilities. While scaling capacity in allied hubs cannot eliminate site-specific hazards, provided capacity is distributed across multiple sites, it offers redundancy that dampens the impact of any single local shock. This means bolstering the existing industrial base in trusted countries to reduce fragility while maintaining alignment on geopolitical and normative priorities. In contrast to classical friendshoring, which emphasizes relocating supply chains to safer jurisdictions [39], this approach reinforces the current supplier base

through targeted investment.

One promising vehicle for doing so is co-investment through a SWF. SWFs have increasingly pursued co-investments to build strategic relationships, particularly with partners who share political and ideological alignment [40]. A US SWF could secure access to these tools and help suppliers scale more rapidly by supporting the expansion of semiconductor-related infrastructure in countries like Japan and the Netherlands. In the longer term, such co-investments provide opportunities for long-term learning and domestic capability development, gradually reducing dependencies. The United States has already applied a similar logic in the raw materials sector. Under the Minerals Security Partnership, the Department of Defense committed \$120 million to a rare earth separation facility in Texas using inputs from Australia [41]. The United States and Australia further agreed to finance new mineral projects jointly. A parallel structure could be developed for semiconductor tooling and other upstream compute inputs.

Domestic Energy Production

Securing US leadership in frontier AI development will require unprecedented investment in physical infrastructure. Some have estimated that training clusters alone could cost up to \$1 trillion and consume over 20 percent of US electricity by 2030 [42]. Without a decisive domestic buildout, authoritarian regimes may come to host and control the infrastructure essential to advanced AI development, threatening both competitiveness and security. A separate projection places global pre-revenue investment in compute infrastructure at up to 10 percent of world GDP in anticipation of TAI [22]. One key barrier is uncertainty: when the path from compute spending to concrete automation outcomes is unclear, even substantial long-term returns may not be enough to overcome investor hesitation. This ambiguity in the return profile of AI infrastructure may slow capital allocation and inhibit progress at a critical inflection point.

A SWF could play a role in resolving this coordination problem. SWFs are well-positioned to take on long-horizon, high-risk investments due to their insulation from short-term market pressures and their mandate to serve national strategic interests, resulting in higher risk tolerance than traditional institutional investors [43]. By prioritizing long-term returns, SWFs such as Singapore's Temasek, CIC, and the Korea Investment Corporation have acted as a form of "patient capital," a model of investment that has historically supported technologies that require extended support, including GPS, the internet, and RFID [44]. Although many SWFs remain globally diversified, there is a growing trend toward strategic domestic investment. Several funds now direct capital to local sectors aligned with national priorities, including healthcare, cybersecurity, and biotechnology, with the goals of strengthening national innovation capacity and crowding in private investment [45]. A US strategic fund could extend this approach to energy infrastructure.

AI development is power-intensive and increasingly constrained by electricity availability [46]. To support the next generation of US-based AI data centers, energy investments could prioritize firm, scalable, and reliable sources—including natural gas, geothermal energy, small modular nuclear reactors, and existing large nuclear plants—to guarantee a stable energy supply for compute clusters [47]. By absorbing front-loaded risk through equity stakes in strategically important energy projects, a SWF could play a catalytic role by attracting private investment to sectors where capital is scarce or risk is high, thereby crowding in actors who might otherwise remain on the sidelines [48]. The fund could also consider investing through publicly traded companies to enhance transparency and credibility, a strategy that, under the logic of “legitimacy and decoupling”, allows SWFs to quietly pursue national strategic objectives while outwardly conforming to financial norms [33].

Security Infrastructure

Protecting advanced AI systems, particularly those with internet access, against highly capable threat actors will require a significant upgrade in security infrastructure, which may take years to develop and deploy [25]. Without proactive planning, these safeguards are unlikely to materialize in time. Promising approaches include restricting physical bandwidth between sensitive systems and external networks, designing hardware that enables inference without exposing model weights, and creating fully isolated environments for training and high-stakes research.

A SWF can anchor early investment in safety-critical technologies, accelerating their adoption and helping to establish them as industry defaults. While most economic models of increasing dominance focus on firm-level competition, the underlying mechanism, where early advantages in investment compound over time, can also apply to state actors. When returns to investment increase with a firm’s state (e.g., a larger user base, lower production costs, or greater trust from customers and partners), early investments can reinforce dynamics that lead to market dominance [49]. If a SWF targets technologies with similar feedback loops, it can simulate a lead firm’s role and shift the entire ecosystem’s trajectory.

For example, a US SWF could invest in hardware mechanisms like FlexHEGs, a proposed hardware mechanism integrated into AI chips to enforce agreed-upon rules about AI development and usage, allowing governments or international coalitions to set flexible, auditable limits without needing constant external oversight or access to sensitive information [50]. By backing such technologies early, a SWF could reduce coordination costs, de-risk adoption, and encourage downstream firms to build around secure-by-default infrastructure. If successful, this would raise the baseline for security across the AI stack.

Equity in Frontier AI Companies

Frontier AI companies often face incentives that may inadvertently compromise national security, such as prioritizing speed over safety in development [51]. These risks are amplified by the growing role of foreign SWFs in shaping the sector. Several Gulf states are rapidly expanding their AI investments as part of national strategies like Saudi Vision 2030 and the UAE National AI Strategy 2031, including direct stakes and partnerships, such as the UAE's Stargate Project with OpenAI and Oracle [52]. Additionally, while US Executive Order 14110 sought to mitigate such risks by requiring federal agencies to enforce safeguards and reporting requirements for powerful AI systems [53], reliance on bureaucratic enforcement, inter-agency coordination, and sustained political will can make regulation vulnerable to delay and rollback. Indeed, the order was rescinded by a new administration in early 2025 [27]. In this context, minority equity stakes that embed strategic control rights offer an alternative, potentially more resilient lever for influencing the behavior of frontier AI companies.

Shareholder activism refers to the practice of SWFs using their ownership stakes in companies to influence corporate behavior and governance [33]. Activist shareholders may vote on key issues, press for strategic changes, or advocate for stronger management oversight. For example, Norway's GPFG promotes transparency and ethical standards in portfolio companies [33], and Temasek actively monitors corporate governance to reduce agency problems and improve firm performance [54]. These actions can be categorized as proactive governance, such as seeking board influence or advancing long-term strategy, and reactive governance, such as blocking specific transactions [55]. While most shareholder activism occurs in public markets and for financial motivations, the underlying principle can extend to private firms for strategic purposes. Governments could influence private companies through less visible but meaningful strategic equity stakes. This is particularly relevant for frontier AI companies, which include OpenAI, Google DeepMind, Anthropic, xAI, and Meta [56], most of which are not publicly traded, as shown in Table 1.

Table 1: Public Trading Status of Frontier AI Companies

AI Company	Country	Publicly Traded?
Anthropic	USA	No [57]
Google DeepMind	USA and UK	No [58]
Meta	USA	Yes [59]
OpenAI	USA	No [60]
xAI	USA	No [61]

Minority equity positions in frontier AI companies may enable the government to negotiate preferred-share terms that embed negative control powers. These powers could prohibit changes to IP ownership or data center location. This creates an internal brake on harmful strategic moves, working faster and more

quietly than post hoc national-security reviews, like those conducted by the Committee on Foreign Investment in the United States (CFIUS) [62]. Side letters—contractual agreements to grant bespoke rights to minority investors—could be used to secure reporting obligations, most favored nation clauses, or vetoes over sensitive governance actions without formal control [63]. Unlike formal rulemaking, ownership stakes could offer a mechanism to access information, shape model deployment standards, and influence development priorities. This approach could provide faster, more stable levers for steering AI development than relying solely on regulatory channels. Golden share structures are especially attractive. Initially used in the UK during privatizations to protect national interests, a golden share grants special ownership rights, including veto power over certain decisions [64]. Unlike controlling stakes or full nationalization, which may trigger backlash or accelerate race dynamics, exacerbating the risks of conflict, control loss, and institutional breakdown [65], they can preserve private-sector activity while giving the state veto rights over a set of strategic decisions.

Although SWFs have historically taken care to avoid triggering political backlash, often by steering clear of majority ownership or sensitive sectors to counter concerns about espionage or strategic influence [37], recent patterns highlight a potential shift. Some governments, all from autocratic regimes, may now be positioning themselves to gain strategic footholds in the development of frontier AI through the acquisition of stakes, as shown in Table 2.

Table 2: Sovereign Wealth Fund (SWF) Investments in Private Frontier AI Companies

Country	Regime Type	SWF	AI Company
Oman	Closed autocracy	OIA	xAI [66]
Qatar	Closed autocracy	QIA	xAI [67]
Saudi Arabia	Closed autocracy	KHC	xAI [68]
Singapore	Electoral autocracy	Temasek	OpenAI [69]
United Arab Emirates	Closed autocracy	MGX MGX Mubadala	OpenAI [70] xAI [71] Anthropic [72]

Note: Based on the “Regimes of the World” typology [20], using data from the V-Dem v15 dataset [19].

As valuations rise and capital inflows grow, individual equity stakes carry diminishing influence, particularly if foreign funds acquire earlier or larger positions. The relative distribution of ownership may shape whose preferences dominate key decisions. Earlier entry may allow democratic governments to secure influence before those opportunities narrow.

Given AI’s potential role in reshaping economic power, military capabilities, and societal governance, such positioning could carry consequences far beyond those associated with past investments. Authoritarian-controlled SWFs pose a strategic risk when they acquire influential stakes in critical infrastructure, as seen in Eurozone banks, due to their potential to project

economic and political influence beyond national borders [73]. This risk is compounded by the opaque governance structures of many authoritarian SWFs, which lack public accountability and are often tightly controlled by ruling elites [74]. When such funds invest in frontier AI development, their motives may extend beyond commercial interests, including regime preservation, geopolitical leverage, and soft power projection. The absence of institutional checks makes these investments difficult to scrutinize, amplifying their systemic risk. In this context, ensuring that control remains anchored in democratic jurisdictions may be an important safeguard against the misuse of advanced AI systems for illiberal purposes, especially given the risk of long-term institutional lock-in.

Public Ownership of Transformative AI

Economic Scenarios

Frontier AI systems are demonstrating rapid advances in high-skill cognitive tasks, raising implications for the automation of the economy. A large-scale empirical analysis of over 4 million Claude.ai conversations found that AI is already used for at least 25% of tasks in 36% of occupations, with 43% of use cases suggesting outright automation rather than augmentation [75]. In parallel, AI accuracy on advanced benchmarks has sharply improved over the past year, with scores rising by 18.8 and 48.9 percentage points on the Massive Multitask Language Understanding and Graduate-Level Google-Proof Q&A benchmarks, respectively, and jumping from 4.4% to 71.7% on the SWE-bench coding test [18]. These trends suggest that AI is approaching human-level competence in select areas of complex problem-solving. Such trends have prompted warnings from leading AI companies. Anthropic CEO Dario Amodei suggested that up to half of all white-collar jobs may be automatable soon [76], and OpenAI CEO Sam Altman has projected that AI agents could soon function as scalable “virtual coworkers” capable of performing tasks at the level of junior professionals [77].

Korinek and Suh [78] have outlined two possible trajectories for the future of AI-driven automation. In the unbounded scenario, AI continues to automate increasingly complex tasks, but humans retain a comparative advantage in ever more sophisticated domains. In contrast, the bounded scenario assumes that human cognitive tasks plateau in complexity; once AI systems surpass this threshold, they could automate the entirety of human labor. In both cases, the economic impact hinges on the relative pace of automation and capital accumulation. If automation proceeds gradually and is matched by investment, productivity gains may translate into higher wages and sustained demand for labor. However, if AI capabilities outpace capital growth or human tasks are ultimately bounded, wages may decline before full automation is reached. As AI substitutes for labor, the supply of human work becomes increasingly abundant and less valuable. This trend could be further accelerated if AI automates R&D, potentially triggering explosive technological growth in which human labor becomes broadly obsolete.

These wage pressures scale up to reshape macroeconomic dynamics. As AI substitutes for labor, the labor share of national income declines while capital absorbs a growing share of gains [79]. AI models, unlike human workers, can scale rapidly through compute investment rather than population growth [22], increasing the risk that economic power accrues to those with control over

these systems and becomes decoupled from broad-based participation [80]. In this context, the critical question becomes who holds the capital. As AI scales, its gains may accrue disproportionately to private investors, corporate stakeholders, and SWFs with equity in leading AI companies. The distributional outcomes of automation thus turn not just on its technical trajectory but on the ownership structure itself.

TAI could emerge under a closed or open-source development regime, with different consequences for control and public benefit. State ownership may become less relevant in an open-source scenario where model weights and code are freely released. Open-sourcing highly capable models significantly weakens any single actor's, including governments', ability to maintain ownership or enforce controls, as public release of weights and architectures limits the means to monitor use or prevent proliferation [81]. By contrast, in a proprietary development context, where access and control are concentrated in the hands of a few actors, ownership becomes an important lever for the state to pursue public interest objectives. However, the relevance of these ownership pathways may diminish if AI benefits prove large and diffuse enough to distribute widely through market forces alone, potentially making formal ownership structures less critical for ensuring broad public benefit [82].

This raises the question of which ownership levers are most strategic. Compute requirements multiply approximately fourfold per year, and compute availability will likely remain a key constraint on near-term AI capabilities [83]. However, compute is fundamentally an enabler of capability, not a durable point of value capture. Once a model is trained, it is a non-rival asset that can be cheaply copied and scaled, subject to inference costs, meaning that value bottlenecks shift to model ownership and control. Over time, compute itself may become commoditized or cease to be the primary constraint on AI progress. Hardware costs have already declined by two orders of magnitude from 2000 to 2022 [84], and efficiency gains or breakthroughs in model architecture could further reduce dependence on scarce compute. Some researchers anticipate a potential "compute overhang" scenario, where improvements in software enable rapid exploitation of abundant, underutilized hardware [85]. Even if compute remains valuable, companies may increasingly vertically integrate their compute supply chains, as illustrated by moves from Google, Microsoft, and Nvidia to align chip design, cloud infrastructure, and model development [86]. In such an environment, targeting compute ownership may offer diminishing leverage over AI value capture relative to targeting ownership of the companies that develop and control transformative models themselves.

For these reasons, this paper highlights public ownership stakes in frontier AI companies as a more strategic lever than compute infrastructure for shaping the long-term distribution of benefits from TAI in a world where TAI remains proprietary. At the same time, compute remains a critical enabler and a valid target for SWF investment. Therefore, this paper outlines a spectrum of possible state relationships to the ownership of a proprietary TAI model and

examines the implications of different levels of involvement and control. Because SWFs enable states to take equity positions across a broad spectrum of ownership levels [87], they are examined within this analysis to assess how their leverage may shape TAI ownership. While all scenarios are examined, the analysis suggests that minority stakes via a SWF may offer a particularly feasible pathway for states to capture a share of the economic value created by TAI.

Ownership Pathways

No stake

In one scenario, the state holds no formal ownership stake in the entity developing TAI. Once such a system is deployed, the state assumes a passive role, dependent on voluntary cooperation, negotiation, or geopolitical leverage to secure access or influence. While states may attempt to exert oversight through regulation, these mechanisms may be constrained by jurisdictional boundaries and institutional capacity. Multinational corporations with significant influence can often wield substantial influence over policymakers through actions like lobbying [88]. If the state does try to exert more influence, firms motivated by profit maximization may relocate to more permissive regulatory environments, further limiting the effectiveness of domestic governance. This form of regulatory and jurisdictional arbitrage can be seen in contemporary examples, such as the exodus of cryptocurrency businesses from New York following the introduction of BitLicense or the relocation of firms from the EU after the implementation of AMLD5 in pursuit of more favorable regulatory environments [89]. Should a foreign state achieve monopolistic control over the TAI model, it may induce economic dependency among other states. The extent and continuity of access would be contingent on the controlling state's political interests and relationships, which are inherently dynamic over time.

If a firm remains physically and legally anchored in the domestic jurisdiction, the state may retain additional avenues for influence. Governments may compel private firms under emergency statutes such as the Defense Production Act of 1950, which was used extensively during the COVID-19 pandemic to prioritize contracts, allocate medical resources, and expand vaccine production [90]. While these tools can exert powerful short-term control, they are episodic, politically contingent, and offer no enduring claim on AI-generated wealth or governance rights. If current economic systems survive during the economic transition, corporate taxation may allow states to obtain a portion of the economic value TAI produces. There may also be scenarios in which the state can negotiate access or have royalty deals, especially if these are contractually made before the advent of TAI. In other cases, companies may adopt softer precommitments that signal alignment with public benefit but are voluntary and largely unenforceable. For example, Anthropic is a Public

Benefit Corporation committed to serving humanity’s “long-term benefit” [91]. While this signals intent, it does not guarantee how the company will act if it develops TAI. Similarly, the Windfall Clause proposes that AI companies pledge to donate a portion of profits if they achieve windfall-level returns [92]. Though such mechanisms aim to share benefits with society, they do not confer control rights to the state and may be difficult to enforce, making their long-term effectiveness uncertain.

Whether the state should own stakes in frontier AI companies can be examined through the lens of residual control rights and incomplete contracting theory. When contracts cannot account for all future contingencies, especially those involving societal-scale risks, ownership becomes a crucial determinant of who has decision-making authority under uncertainty [93]. In the case of TAI, private firms may prioritize efficiency or profit in ways that fail to capture non-contractible public concerns. The inherent unpredictability of TAI’s societal impacts make it difficult for states to guarantee public outcomes through regulatory contracts alone. Without some degree of ownership, the state may struggle to ensure alignment with public interest or manage macroeconomic risks. Moreover, if TAI leads to extreme wealth concentration among a few stakeholders, public ownership may help prevent democratic erosion or economic inequality. SWFs, in particular, offer a viable mechanism for capturing some of the upside value of AI for broader societal benefit.

Minority stake

A modest ownership stake in a leading AI company could provide a state with substantial fiscal capacity to pursue national objectives, because the potential economic and strategic returns from TAI are immense [22][94]. As such, it is in a state’s strategic interest to secure some degree of ownership in firms developing frontier AI systems. However, most of these companies are concentrated in a few countries and are privately held [18]. Acquiring a controlling stake is neither politically feasible nor financially realistic for most states. This makes significant minority ownership a more practical route to ensure public alignment in the development and deployment of TAI. Minority interest, typically defined as less than 50% of voting stock [95], does not confer full control but can still provide strategic influence. Research shows that managerial control can be exercised with far less than a majority stake. Firm value follows a non-linear pattern: it increases with board ownership from 0 to 5%, declines between 5-25% due to entrenchment, then rises past 25% [96]. This suggests that meaningful influence is often achieved well below majority thresholds. Even state-owned enterprises (SOEs), which frequently invest with strategic intent, tend to pursue minority stakes rather than full ownership, especially when targeting large or high-value firms [97]. In cases where a single investor holds the most significant individual stake, even a minority position can provide board representation and influence over corporate governance [98].

Failing to secure such a stake risks ceding strategic control over what may become the most powerful economic tool. Relying entirely on foreign firms or private domestic actors, without any public ownership, exposes states to significant risks. Foreign governments have no obligation to share TAI's benefits and may treat it as a strategic asset. Even in democratic contexts, private shareholders may not prioritize global equity or public goods. Indirect exposure, such as through pension funds holding shares in intermediaries like Microsoft, which owns 49% of OpenAI [99][100], may be vulnerable to shifts in corporate governance. Moreover, in foreign jurisdictions, such holdings can be diluted, stripped of voting rights, or even expropriated, especially during geopolitical tension or legal ambiguity. Latin American precedents show that foreign investors are often subject to coerced renegotiation or informal expropriation without meaningful recourse [101]. Additionally, what seems like a stable alliance today may not hold in the future: relying on foreign-controlled entities for transformative technologies risks entrenching dependencies.

SWFs offer a viable and legitimate channel through which states can acquire minority stakes in AI companies, especially since acquiring minority stakes is a common strategy among SWFs globally [102]. Several SWFs have already begun building exposure to frontier AI companies despite their private status and limited accessibility, as shown in Table 2. Strategic entry opportunities may be both time-limited and relationship-dependent. As generative AI capabilities become more evident, valuations will likely rise sharply, illustrated by Nvidia's recent surge following an unexpected earnings report tied to AI demand [103]. Because it is impossible to predict which lab may first achieve TAI, sovereign investors could avoid concentrating bets and instead pursue diversified minority positions across multiple frontier AI companies. In line with principles of the social contract [104], such investments can be seen as part of the state's responsibility to safeguard public welfare in the face of disruptive technological change. However, even with capital and legitimacy, access is not guaranteed. Some firms may reject sovereign capital on political or national security grounds. For example, Anthropic excluded Saudi Arabia's SWF from a funding round over such concerns [105].

Majority stake

A controlling interest is more than 50% of a company's voting stock [Smith, 2021]. Holding the majority of a firm's equity confers decisive voting power and direct influence over strategic decisions. This could be achieved by creating an SOE at the outset or through nationalization. The feasibility of majority control depends heavily on timing: establishing public ownership from the outset is likely less contentious than intervening after TAI becomes economically or geopolitically significant. Post-hoc nationalization may provoke legal challenges, capital flight, or diplomatic retaliation. For example, Iran's nationalization of the Anglo-Iranian Oil Company in 1951 led to a British-imposed embargo and eventually a coup supported by foreign powers

[107]; Chile's 1971 copper nationalization, which retroactively deducted "excess profits" from US firms without compensation, triggered legal retaliation and trade disruptions by multinational corporations and US diplomatic pressure [108]; and Mexico's 1982 bank nationalization led to significant capital outflows and exacerbated the country's debt crisis [109].

SOEs and the nationalization of private companies have been discussed in the context of frontier AI development. The potential for majority state ownership in the context of TAI likely only applies to major powers developing frontier AI models, most notably the United States, with China coming in second [18]. Some have argued that frontier AI development will soon become a matter of national security for the United States, forcing the federal government to take direct control, much like with the Manhattan Project during World War II [42]. This perspective is reinforced by the US-China Economic and Security Review Commission, which, in its 2024 Annual Report to Congress, advocated for a Manhattan Project-style initiative to pursue artificial general intelligence as a strategic priority in the competition with China [110]. In the case of China, the government could plausibly nationalize frontier AI companies operating within its jurisdiction, such as DeepSeek. The Chinese Communist Party has a track record of embedding internal Party organizations within state-owned and private firms to ensure alignment with state objectives [111]. Moreover, China retains the sovereign right to expropriate private assets if deemed in the national interest [112].

Empirical data suggest that less than one percent of SWF investments exceed the 50% threshold required for obtaining a controlling interest [113]. While most SWFs traditionally favor passive investment strategies, favoring debt instruments or minority equity positions, some have demonstrated increasing flexibility in recent years, with instances of SWFs acquiring controlling interests through corporate acquisitions [114]. The CIC, for example, holds controlling stakes in most major Chinese financial institutions and has acquired significant positions in foreign firms [115]. While many SWFs have formal or informal guidelines against hostile takeovers or control-seeking behavior [116], these policies are not immutable. In jurisdictions that house frontier AI companies like the US, national security concerns may lead governments to block or closely scrutinize foreign investments. For example, the United States uses the CFIUS to review and potentially prohibit investments by SWFs or other foreign entities in sensitive sectors like critical infrastructure [117]. Given these restrictions, states that do not house frontier AI companies may find that acquiring full control is politically infeasible, making minority stakes, golden shares, or contractual side agreements a more realistic path to securing public ownership of TAI.

Public Management of AI-Derived Wealth

A central question in the political economy of TAI is what role the state will play in owning or controlling AI-derived resources. While such ownership offers a path to more equitable outcomes, it does not guarantee social or economic benefit. Without deliberate policy mechanisms like SWFs, state control over TAI-related wealth could perpetuate inequality and fuel political or economic instability. As automation advances and the potential for wage collapse grows, the challenge becomes how to allocate the resulting surplus in ways that preserve economic security. This concern is especially acute if the benefits of AI fail to diffuse widely through market mechanisms [82]. In such a context, the state's role should extend beyond regulation to include the active design of institutions that distribute AI's gains. States that acquire equity in TAI-producing firms will face choices about whether to let market forces operate unchecked, direct returns toward public services, or reimagine ownership structures altogether. The remainder of this section contrasts two broad approaches, *laissez-faire* and public return, and assesses their implications through analogies such as the resource curse. It explores how unchecked control over rent-generating assets can entrench elite power and how tools like savings and stabilization funds might mitigate these risks and help ensure that the benefits of automation are shared more equitably.

Laissez-Faire

One possible scenario is that the state retains control over the bulk of productive capital associated with TAI while no longer requiring labor from most citizens. In such a setting, wealth becomes increasingly concentrated, yet public welfare provision declines due to neglect or deliberate policy choices. History offers no enduring examples of elites hoarding capital, abandoning labor, and withholding basic welfare without triggering social upheaval or humanitarian crisis. For example, the *laissez-faire* policies that shaped Britain's response to the Great Irish Famine (1845–1851) illustrate how adherence to market principles over humanitarian aid can produce catastrophic outcomes. During the famine, many Irish landlords, seeking to maximize profits, evicted tenants from small subsistence plots in favor of larger holdings suitable for pastoral farming [118]. These clearances, carried out amid mass starvation, left much of the rural poor homeless and destitute, contributing by the early 1850s to over 1 million deaths and 1.5 million emigrations from a population of 8.1 million. A similar pattern of elite hoarding and policy neglect was evident during North Korea's "Arduous March" in the 1990s. As the

country's centrally planned economy collapsed, the Public Distribution System—the mechanism responsible for food rationing—disintegrated [119]. Rather than pursuing reforms or accepting international assistance, the regime responded by selectively distributing aid to politically loyal regions while allowing disfavored areas to starve, resulting in a famine that claimed between 600,000 and 1 million lives. Despite mass suffering, the regime remained in power, resisting both liberalization and decentralization even as citizens turned to foraging and black markets to survive.

These cases underscore the idea that when governments control vast wealth but sever their obligations to the broader population, the outcome may be catastrophic. In the case of TAI, the risks may be further magnified if these systems enable the automation of political control through surveillance or propaganda, reducing the ability of citizens to resist even under conditions of deprivation. Modern authoritarian regimes often maintain power not through ideology or brute force alone, but by manipulating perceptions of competence, co-opting elites, and controlling access to information [120]. In a TAI-enriched state, surplus wealth could be used not to improve citizens' lives, but to entrench elite dominance through media control, censorship, or patronage.

The Resource Curse

TAI presents a political economy dilemma similar to the traditional resource curse—the paradox whereby countries rich in natural resources often experience slower development, institutional decay, and heightened political instability [121]. Some have referred to this analogy as the “intelligence curse” [122], drawing a parallel between the effects of oil wealth and the dynamics that could accompany the control of advanced AI systems. Just as windfalls from non-renewable resources, particularly when state-owned, have historically reinforced authoritarian regimes, fostered corruption, and weakened institutional accountability [123], the centralization of TAI revenue streams may produce similar or greater distortions in states with weak institutions or authoritarian governance.

At the core of the resource curse lies the challenge of unearned income: when states acquire wealth without the mediating force of taxation or market participation, the incentives for accountability and long-term investment are often undermined [124]. Resources that generate concentrated, easily captured revenue streams are particularly susceptible to this dynamic, especially in states where institutions are weak or underdeveloped. In countries where strong democratic institutions, transparent bureaucracies, and the rule of law are already in place, resource wealth is more likely to be managed in ways that support inclusive growth and intergenerational equity. For example, when Norway discovered oil in the late 1960s, it already had a stable democratic system and low levels of corruption [125]. These foundations enabled the Norwegian government to implement long-term policies, including creating a SWF, to mitigate volatility and ensure broad social benefits from oil revenues.

Similarly, Chile established a stabilization fund to manage copper revenue volatility, drawing on independent expert panels and a rules-based fiscal framework to maintain budget discipline during shocks [126]. By contrast, countries without these preexisting institutions often cannot resist the distortive effects of resource rents. Nigeria, for instance, earned hundreds of billions of dollars in oil revenues over several decades, yet saw little improvement in per capita income or human development indicators, and instead saw an increase in rent-seeking and weakened accountability mechanisms [127]. Equatorial Guinea's oil boom entrenched authoritarian rule, enabling a small elite to monopolize wealth, and exacerbated ethnic divisions [128]. The regime directed funds toward personal enrichment and symbolic state projects rather than using resource revenues to fund health, education, or infrastructure. At the same time, the broader population remained impoverished and politically marginalized.

These patterns provide a lens through which to assess the political economy of TAI. Like oil, AI systems may generate highly centralized, state-capturable rents. Governments can secure control and resist institutional reform if access to high-capability models is monopolized. In extreme cases, the centralization of AI capabilities raises the risk of an AI-enabled coup, wherein exclusive access to powerful, secretly loyal models could permit a small group to subvert state institutions [129]. As with traditional rentier economies, governments that secure such assets may substitute politically unaccountable revenue for tax-based financing, reducing incentives to be responsive to citizens' needs. This mirrors dynamics in oil-rich rentier states, where non-tax revenues weaken fiscal capacity and the social contract [130][131]. In these settings, state authority becomes decoupled from citizen consent, granting broader discretion over windfalls and locking in institutions that sideline private investment and broad development [132][133]. In China, where income tax is limited, fiscal policy is weakly tethered to public accountability, whereas Canada's broad, progressive tax base underpins a more participatory democratic structure [134]. AI-generated rents may also fuel populism, with regimes deploying windfalls for short-term political gain, a logic akin to petro-populism [135].

States currently lack the institutions to manage TAI's economic and political consequences. Without proactive governance mechanisms, the centralization of AI-generated wealth risks reproducing similar extractive and exclusionary dynamics in traditional resource curses. By investing for long-term intergenerational benefit, SWFs have been used to combat the resource curse, and could similarly help states avoid the intelligence curse.

Public Benefit Models

Rather than accumulating capital, states can use public assets to support citizens directly through income transfers or indirectly by strengthening services and stabilizing economic conditions. This section explores how public capital can be deployed through savings and stabilization funds to ensure

widespread economic and political security during and after transitioning to an AI-automated economy.

Savings funds

Traditionally, savings funds have been used as long-term tools for national financial planning, often supporting future obligations like pensions [11]. However, the rationale for saving this way may shift in an economy shaped by TAI. If labor becomes largely automatable and traditional employment diminishes, the need for pensions, rooted in the distinction between working and retired populations, may become obsolete. In such a context, the emphasis may need to move from saving toward distribution, ensuring that citizens continue to share in national wealth irrespective of employment status.

Universal Basic Income

One policy proposal that aligns with this vision is the direct distribution of resource revenues to citizens, followed by taxation. This approach has been suggested as a way for governments to restore public trust, increase tax compliance, and discourage corruption by strengthening the accountability link between citizens and the state [136]. It also helps mitigate the resource curse by shifting state-citizen dynamics from top-down patronage to mutual responsibility [137]. A SWF could therefore be repurposed to distribute dividends directly to citizens as a form of non-labor income. While many existing SWFs have historically resisted such distribution models, often citing concerns about consumption, dilution of returns, and savings erosion, empirical studies of Alaska's Permanent Fund Dividend (PFD), the only SWF that distributes a portion of its returns directly to residents, challenge these objections [138]. Alaskan recipients commonly save or use their dividends for debt reduction, and the fund has continued to grow even while distributing over half its income since the 1980s. Though initially perceived as a one-off windfall, the PFD has become a consistent part of personal income, shaping consumption patterns and supporting roughly 7,000 jobs and \$1.1 billion in economic activity annually. In some parts of Alaska, it accounts for up to 13% of per capita income [139]. These dividends are also subject to federal income tax for adults [140]. Unlike conventional universal basic income (UBI) models that rely on redistributive taxation, the PFD represents a non-redistributive form of income: it treats natural resources as common property and returns economic rents directly to the public [141][142].

UBI has already been proposed as a solution to address income displacement caused by AI and automation [143]. Increasing the public share of AI capital profits can significantly reduce the level of AI-driven productivity needed to make UBI financially viable [144]. This effect is strongest during early deployment, when public capture can sharply lower the threshold, though returns diminish at higher capture levels. These dividends could also be subject to federal income taxes, creating a cycle that supports individual incomes and

reinforces the tax base and state accountability. The share of SWF returns allocated for distribution would depend on factors such as the level of public capture of TAI-generated wealth, the resulting size and return profile of the fund, demographic pressures, and the scale of labor displacement.

Universal Basic Assets

As an alternative or complement to UBI, the concept of Universal Basic Assets (UBAs) has been proposed for a future where AI automates a large share of economic activity. UBAs emphasize broad-based ownership of productive capital, offering a long-term solution beyond income support to focus on wealth-building and economic agency [145]. Unlike UBI, which provides ongoing income regardless of asset ownership, UBA models advocate for collective or individual stakes in capital-generating assets. However, no examples exist of SWFs evolving to distribute ownership through collective mechanisms. This absence may reflect the inherent risks: without robust governance, democratizing shares of public wealth can lead to unintended consequences. Historical precedent shows the dangers of poorly structured asset democratization. During Russia's 1990s voucher privatization program, efforts to quickly and equitably transfer state assets to the public backfired: wealthier and better-informed individuals were able to purchase vouchers from the poor at low prices, leading to a rapid reconcentration of ownership [146]. The unrestricted transferability of shares enabled a small elite to consolidate control, undermining the program's goals.

If similar approaches were applied to AI-related capital, we might witness comparable patterns of upward wealth concentration. To prevent this, any model of citizen ownership would likely require safeguards, such as restrictions on liquidity or transferability, to avoid exploitation. Handing out equity without protections risks reinforcing the inequalities that such models aim to address. Governments or contracted asset managers manage most SWFs and are rarely subject to direct citizen input or oversight [147]. To support the idea of a UBA, an option would be to structure a SWF as a public trust. Instead of disbursing individual shares, collective ownership could be defined in the mandate. In this model, the fund would exist exclusively for the benefit of current and future citizens, with legal protections ensuring that its assets are used solely for such purposes.

Stabilization Funds

Stabilization funds could play a role in managing economic volatility during the transition to widespread automation, especially as AI accelerates change. While automation promises long-term productivity gains, it does not ensure smooth or immediate economic adjustments. Rapid technological shifts can produce significant transitional shocks, particularly when the pace of automation surpasses the development of new job categories or exceeds the capacity of institutions to retrain displaced workers [80]. Assuming seamless adaptation risks underestimating short-term unemployment, while assuming

permanent displacement may overstate the lasting impact [22]. Historical precedent underscores this point: the British Industrial Revolution saw rising productivity coincide with decades of stagnant wages and a falling labor share, a period known as "Engels' pause" [80]. Stabilization funds could capture windfalls from TAI and redeploy them during downturns, cushioning labor markets and public finances from volatility.

Stabilizing Labor Markets

Stabilization funds, often used to buffer commodity-driven economies from volatile price cycles, could be repurposed to navigate the economic turbulence brought by TAI. Chile offers a useful precedent: heavily reliant on copper exports, the country established the Economic and Social Stabilization Fund in 2007 to manage revenue windfalls and insulate public spending from commodity shocks [148]. The fund proved crucial during the 2008 financial crisis, enabling Chile to maintain fiscal stability despite collapsing copper prices [10]. As AI transforms labor markets, a similar mechanism could help governments absorb shocks, redistribute benefits, and support displaced workers in real time.

Rather than relying solely on reactive stimulus measures, as seen in the US CARES Act during the COVID-19 pandemic [149], a dedicated stabilization fund for AI-era disruptions could offer more structured and anticipatory support. For example, it could provide supplemental income to sectors disproportionately impacted by automation, functioning in tandem with or as a replacement for programs like Trade Adjustment Assistance, which supports retraining and relocation for workers harmed by trade exposure [150]. Embedding automatic stabilizers into these funds could strengthen countercyclical responses without the delays of new legislation by automatically increasing benefits or reducing tax burdens as economic conditions deteriorate [151]. In the context of fast-moving AI disruption, such features would allow stabilization funds to respond immediately, boosting income support as jobs disappear and lightening tax loads as household earnings decline. Additionally, workers displaced by AI performing qualitatively different tasks may require targeted education and reskilling pathways to transition effectively, as the absence of such support could deepen inequality and reduce productivity [80]. This view is echoed in recent research arguing that fiscal policy should prepare for gradual and highly disruptive AI scenarios by strengthening unemployment support and investing in upskilling and reskilling programs to cushion transitions and reduce inequality [152]. Stabilization funds could finance these efforts on an ongoing, rather than emergency, basis.

Stabilizing Government Revenue

As TAI reshapes the economy, governments may need to decouple fiscal spending from volatile revenue streams. Stabilization funds can buffer public budgets against the swings in economic output and tax revenues that may

accompany rapid automation, helping governments maintain public services and economic stability. Just as Chile's reliance on copper, at times contributing up to one-third of total fiscal revenue, exposed its government to commodity price shocks [148], states may find themselves similarly dependent on the economic surpluses generated by TAI. If these revenues prove cyclical or concentrated in specific sectors, boom-and-bust dynamics could destabilize public finances. In moderate downturns, such volatility might prompt increased debt issuance. However, fiscal instability could threaten the state's legitimacy in more severe cases. Historical experience offers a warning: in Weimar Germany, economic collapse contributed to a broader governance crisis, as the state struggled to meet social and economic obligations amidst hyperinflation and falling revenues [153]. This failure undermined public trust and played a significant role in political fragmentation and democratic disintegration [154].

As TAI transforms labor markets and alters the structure of the tax base, particularly if labor income declines while capital and corporate profits rise, governments may find themselves with eroding tax revenues. Rising profits do not automatically translate into public income without robust corporate taxation frameworks. Even firms performing well under automation can contribute little to fiscal stability if their earnings are lightly taxed or easily offshored. If governments cannot deploy adequate countercyclical spending during downturns or if they become over-reliant on highly concentrated sources of revenue, the risk of democratic backsliding and political instability may rise. A stabilization fund could capture surplus revenues during boom periods and smooth expenditures over time, helping insulate public services and welfare spending from these cycles.

Discussion

The case for SWFs as a tool for shaping the trajectory of TAI depends on technical feasibility and institutional durability. Even well-designed funds can falter if the state becomes unstable, governance breaks down, or strategic investment accelerates the dynamics it is meant to contain. This section considers key limitations and risks: the persistence of state institutions, uncertainty around the advent of TAI, the potential for arms race escalation, and the possibility that SWFs entrench elite power rather than serve the public interest. It also examines the structural barriers that could keep some states locked out of the economic gains enabled by TAI.

Institutional Persistence

Rising complexity accelerated by TAI may challenge the continuity of the state as an actor in global governance. Historical analyses suggest that as the returns to managing growing complexity diminish, societies may struggle to sustain their sociopolitical infrastructure [155]. TAI could intensify this challenge by generating rapid economic and social shifts, including labor market disruption, informational instability, and the concentration of wealth and power. Such shifts may overwhelm institutional capacities or erode legitimacy, increasing the risk of systemic dysfunction or collapse.

Even if states endure, the persistence of existing institutions is not guaranteed. Institutional outcomes depend on the interplay between political power, elite economic interests, and expectations about institutional evolution [156]. Institutions tend to persist when they serve the interests of those in power and when changes are perceived as threatening to elite dominance or introducing uncontrollable dynamics. As a result, even economically rational reforms may be blocked if they are seen as politically risky. SWFs, particularly those tasked with managing AI-generated wealth, may become focal points in these dynamics. If wealth and influence are increasingly concentrated among a narrow set of actors, institutional support for SWFs will depend on whether elites view them as tools for reinforcing their position or as threats to it. Elites may resist redistributive uses of SWFs if they believe such moves could trigger democratization or expand state intervention. Conversely, SWFs that help preserve legitimacy or absorb societal shocks may be more politically palatable and more likely to endure.

Historical precedents suggest institutional persistence is possible, especially when institutions adapt to emerging needs. During the Industrial Revolution, the British Parliament reformed rigid property laws to facilitate economic development while maintaining political stability [157]. These reforms allowed for the flexible transfer and repurposing of land, aligning legal frameworks

with changing industrial demands without undermining the institutional order. Designing SWFs with adaptive capacities, such as mandates to address labor disruption or social inequality, may enhance their relevance and legitimacy in an AI-induced transformation. This approach aligns with calls for anticipatory governance structures that can respond effectively to the challenges posed by rapid technological change [158]. Although uncertainty surrounds the long-term trajectories of both states and their institutions, there is reason to expect that SWFs can persist, especially if they evolve alongside the systems they govern.

Uncertainty Around Transformative AI

While this paper explores the potential transformative impacts of AI, it is important to acknowledge the substantial uncertainty surrounding its eventual realization. Central to current optimism in the field is the scaling hypothesis, which suggests that increasing the size of neural networks, the volume of training data, and the amount of compute can lead to qualitative improvements in AI capabilities, potentially without the need for fundamentally new algorithms [159]. Contingent on resolving key bottlenecks such as energy availability, semiconductor production capacity, data scarcity, and network latency, training runs may reach scales of up to $2e29$ FLOP by 2030, around 10,000x the compute used to train GPT-4[83]. However, this trajectory is not guaranteed. Investment and enthusiasm may be driven more by speculative sentiment than by demonstrated performance. It remains possible that either the scaling hypothesis is incorrect or that physical and economic bottlenecks will prevent its full realization. The current enthusiasm for AI may mirror earlier speculative bubbles, such as those surrounding the Dot-Com era or cryptocurrencies, where valuations significantly outpaced underlying technological and commercial maturity [160]. If SWFs make substantial investments in frontier AI companies, a collapse in valuation could lead to public financial losses.

If TAI fails to materialize, a strategic fund focused on AI may resemble a conventional venture capital portfolio, with correspondingly moderate returns. Nevertheless, such a fund could still hold significant strategic value. By taking stakes across compute infrastructure, energy systems, and upstream supply chains, the fund could hedge against shocks and secure access to assets the economy may need in less transformative scenarios. For example, it is estimated that electricity demand in the United States could increase by 81% by 2050 due to widespread electrification, requiring more than a doubling of current generation capacity [161]. Separately, a single frontier AI training cluster could require up to 100 gigawatts of power by 2030—equivalent to more than 20% of current U.S. electricity generation—while total AI-related compute, including inference and multiple clusters, could consume a majority of the national electricity supply within the same timeframe. [42]. In both cases, strategic investment in energy infrastructure is warranted regardless of whether

AI reaches transformative thresholds. These positions may resemble early-stage bets in high-growth technology firms: potentially valuable and well-justified strategically, even if they do not lead to transformative outcomes. For example, In-Q-Tel, the CIA-backed venture capital firm established in 1999, was designed to secure early access to commercially developed technologies and has provided strategic and operational value to US intelligence agencies, even without consistent financial returns [162]. The strategic rationale for maintaining such funds remains intact even if equity ownership in frontier AI companies does not meaningfully contribute to fiscal stabilization or national savings, as SWFs have been justified on grounds unrelated to transformative technologies and may continue to deliver value even if TAI does not materialize.

Arms Race Dynamics

State-led investment in frontier AI development may intensify AI arms race dynamics, especially when funding is directed toward enhancing national security or gaining a strategic advantage. This can trigger action-reaction responses from rival states and escalate technological competition. However, the “AI race” framing itself may be exaggerated, risking a self-fulfilling prophecy that pressures governments and companies to accelerate development further [163]. In this setting, the use of SWFs reflects a rational response to perceived external threats, yet it can also generate heightened insecurity and reduce prospects for international cooperation [164]. Moreover, by channeling substantial public capital into AI development, SWFs may inadvertently incentivize rapid and less cautious innovation, increasing the risk of misuse, accidents, or systems misalignment. Historical analogues and theoretical analyses of AI arms races suggest that strategic or economic imperatives often override safety, raising the possibility of catastrophic outcomes [165].

Governance Risks

SWFs have shown susceptibility to misuse and political interference, as limited transparency and weak oversight have allowed those in power to repurpose public assets for private or political ends [166]. Even when governance frameworks are in place, they often fail to prevent decision-making that prioritizes narrow interests over long-term public value, especially where institutional checks are weak, or fund operations are shielded from meaningful scrutiny. If SWFs are used to acquire ownership in TAI, they may become tools for political entrenchment or misallocation of public wealth, especially if decision-making is concentrated without adequate safeguards. The opacity of many SWFs further compounds these risks, and many SWFs invest through private equity structures exempt from public disclosure requirements [167]. As a result, even in democracies, it may be difficult to trace how public capital is

being used and whether these initiatives align with their stated public purpose. Although the Santiago Principles were developed to promote greater openness and sound governance in SWFs, they fall short of mandating enforceable compliance [168].

Investment Constraints

SWFs are often funded through the budget surpluses derived from commodity exports or the buildup of foreign exchange reserves [169]. However, without such fiscal buffers, some countries have turned to debt financing to grow their sovereign funds [170]. While viable in theory, this pathway is far less accessible for countries burdened by persistent budget deficits, limited natural resource revenues, and high levels of external debt. Debt distress currently affects or threatens more than half of the world's low-income countries [171].

Establishing a SWF requires committing fiscal resources toward long-term investments that may not yield immediate returns. This is particularly difficult for poorer countries, where the marginal utility of public consumption is high, and the opportunity cost of saving is correspondingly steep [172]. In such contexts, allocating funds to a savings or stabilization vehicle may appear politically and economically unjustifiable, especially in the presence of urgent development needs. Frontier AI companies are largely privately held, as shown in Table 1, and investing in them often requires patient capital and a tolerance for illiquidity, traits that are less compatible with the financial constraints of lower-income countries. Even when a state opts to create a SWF, the ability to finance it through debt is limited by sovereign creditworthiness. High debt levels can lead to a debt overhang, where creditors no longer expect full repayment, shutting off access to new financing [173]. While low-income countries face the most acute fiscal constraints, rising public debt and political gridlock have similarly raised concerns about the ability of high-income countries like the United States to fund a SWF [174].

Even when financial hurdles are overcome, legal and political constraints may obstruct meaningful participation in frontier AI markets. For instance, SWFs are less likely to invest in private equity in jurisdictions with strong investor protections, which can reflect deliberate regulatory barriers to foreign state-owned investment [175]. In the United States, the CFIUS is central in reviewing and potentially blocking foreign investments that may pose national security risks, particularly in sensitive sectors like advanced technology [176], presenting a significant barrier for foreign SWFs seeking to acquire stakes in leading US AI companies. Additionally, private firms can independently reject sovereign investment, further limiting access. Corporate governance norms and competition law present additional obstacles: boards may deploy defensive measures like shareholder rights plans to block large-scale acquisitions by sovereign actors, cross-firm investments may trigger antitrust scrutiny, and some companies, like Alphabet, use dual-class share structures that prevent outside investors from gaining meaningful influence or control, even with

substantial ownership stakes [92]. However, in moments of perceived national urgency, states have asserted control over private firms without shareholder consent. During the 2008 financial crisis, the US government acquired an 80% equity stake in American International Group despite shareholder objections [177]. While this intervention occurred during financial distress, legal and corporate resistance may be overridden when strategic interests are sufficiently high. If exclusion from frontier AI is framed as a national security threat, similar mechanisms could be invoked, though doing so outside of crisis conditions may be far more controversial.

If states cannot secure ownership stakes in TAI, they risk being excluded from its economic benefits. This could widen the gap between wealthy and resource-constrained states, reinforcing global disparities. During the 19th century, countries that failed to implement or finance industrial policies amid major technological shifts were largely excluded from long-term economic convergence [178]. A similar dynamic may unfold if states cannot invest meaningfully in the companies that may develop TAI.

Global Benefit Distribution

If some states cannot secure ownership stakes or meaningful participation in the development of TAI, the global distribution of its economic benefits will be questioned. In the absence of inclusive ownership, alternative mechanisms may be needed to ensure that TAI's advantages do not exacerbate existing global inequalities. However, there is considerable uncertainty over whether high-income countries would voluntarily commit to redistributing significant economic gains at the global level. Historically, foreign aid allocations have often reflected the political and strategic priorities of donor countries rather than the developmental needs or policy performance of recipient states, such as former colonial relationships or voting alignment in the United Nations [179]. In the context of TAI, similar dynamics may emerge. States with some degree of ownership of TAI may prioritize strategic and geopolitical objectives when determining how benefits are shared. And if there are benefit-sharing mechanisms, they could be leveraged as tools to solidify alliances, influence global governance arrangements, or extend soft power [82].

Although SWFs are state-owned investment entities and traditionally operate to serve national interests, they could be adapted to support international benefit-sharing goals. However, there are no cases of SWFs redistributing returns directly to other states. Proposals suggest that AI-rich nations could allocate a portion of their TAI-generated wealth to multilateral funds established through voluntary commitments or international agreements, which could then redistribute resources globally [82]. Distribution could occur through two primary channels. First, benefits could be directed to individuals in low-income countries, potentially modeled on existing humanitarian cash transfer programs, such as those administered by the United Nations World Food Programme [180]. Alternatively, funds could be transferred to national

governments, enabling them to implement development programs directly. This mirrors existing practices in global development finance, such as the operations of the World Bank's International Development Association [181].

Conclusion

This paper has examined the potential role of SWFs in mediating the economic and geopolitical dynamics associated with TAI, arguing that they offer states a means of influencing the development, ownership, and distribution of AI-related assets. By enabling long-term investment, facilitating public equity in frontier firms, and supporting mechanisms for post-labor transitions, SWFs may help mitigate the concentration of value and control in advanced AI systems. At the same time, the analysis has highlighted key constraints and risks, including uncertainty around the advent of TAI, governance vulnerabilities within SWFs themselves, and the potential for state investment strategies to reinforce elite interests or exacerbate geopolitical competition.

Because no existing literature addresses the relationship between SWFs and TAI, this paper offers a novel conceptual framework linking sovereign investment to the political economy of AI. While this paper does not argue that SWFs should be the primary mechanism for managing TAI, it proposes that they merit inclusion as one promising tool within a broader portfolio of public strategies for navigating contested frontier AI leadership, accelerating automation, and post-labor economic transitions. In particular, strategic minority stakes in frontier AI companies may provide a practical pathway for states to gain influence and public ownership in AI development. As TAI-generated wealth risks deepening global inequality, it may also be necessary to explore new models for transnational benefit-sharing, including internationally coordinated sovereign investment vehicles that allocate a portion of AI-derived returns toward global public goods. Such mechanisms could be critical to ensuring that TAI serves broader human welfare.

The ultimate effectiveness of SWFs will be contextual: how they are funded, designed, and governed. Further research should identify sustainable funding strategies, assess the viability of different SWF models, and explore the institutional requirements for aligning these tools with long-term public benefit.

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